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Inventor(s)	Hamada; Hiroto et al.

Information processing device

Abstract

The information processing device includes a target unit. When a target object existing in a traveling direction of a vehicle traveling on a road is detected by a detection unit, and it is assumed that a positional relationship between the vehicle and the target object is in a predetermined state, and the target object crosses the road, the control unit displays a target object image showing the target object at a position reflecting the actual position of the target object in the road image showing the road displayed on the display unit in the vehicle.

Inventors: Hamada; Hiroto (Miyoshi, JP), Tsujino; Miki (Nagakute, JP), Hatanaka; Masashi (Toyota, JP), Konishi; Makoto (Nisshin, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)

Family ID: 92195527

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)

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Primary Examiner: Foxx; Chico A

Attorney, Agent or Firm: SoraIP, Inc.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2023-020280 filed on Feb. 13, 2023, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an information processing device.

2. Description of Related Art

(3) WO2020/031912 discloses a technology for enabling an occupant to clearly view a target object present outside a vehicle.

SUMMARY

(4) In the technology of WO2020/031912, target object-related information indicating the target object is displayed on a head-up display (HUD). However, there is room for improvement in terms of display contents of an in-vehicle display for the occupant to grasp the target object present outside the vehicle.

(5) Accordingly, an object of the present disclosure is to provide an information processing device capable of displaying, on an in-vehicle display, contents related to the possibility that a target object will actually cross a road.

(6) An information processing device according to one aspect includes a control unit configured to, when a target object present in a traveling direction of a vehicle traveling on a road is detected by a detection unit, a positional relationship between the vehicle and the target object is in a predetermined state, and the target object is expected to cross the road, display a target object image showing the target object at a position reflecting an actual position of the target object in a road image showing the road and displayed on a display unit in the vehicle.

(7) In the information processing device according to the above aspect, when the target object is detected by the detection unit, the positional relationship between the vehicle and the target object is in the predetermined state, and the target object is expected to cross the road, the control unit displays the target object image at the position reflecting the actual position of the target object in the road image. As a result, the information processing device displays the target object image on the condition that the target object is expected to cross the road. Thus, the contents related to the possibility that the target object will actually cross the road can be displayed on the in-vehicle display.

(8) In the information processing device according to the above aspect, the target object may be expected to cross the road when an actual direction of the target object is a direction toward the road.

(9) In the information processing device according to the above aspect, the target object is expected to cross the road when the actual direction of the target object is the direction toward the road. Thus, the information processing device can determine the possibility that the target object will cross the road based on the actual direction of the target object, and can perform control as to

whether to display the target object image.

(10) In the information processing device according to the above aspect, the control unit may be configured to display the target object image reflecting the actual direction of the target object.

(11) In the information processing device according to the above aspect, the control unit displays the target object image reflecting the actual direction of the target object. Thus, in the information processing device, the occupant can grasp the actual direction of the target object by viewing the display unit.

(12) In the information processing device according to the above aspect, the control unit may be configured to erase the target object image from the display unit when the target object is no longer expected to cross the road after the target object image is displayed.

(13) In the information processing device according to the above aspect, the control unit erases the target object image from the display unit when the target object is no longer expected to cross the road after the target object image is displayed. Thus, the information processing device can reduce the amount of information to be displayed on the display unit when the target object is no longer expected to cross the road.

(14) In the information processing device according to the above aspect, the control unit may be configured to display a vehicle image showing the vehicle at a position reflecting an actual position of the vehicle in the road image.

(15) In the information processing device according to the above aspect, the control unit displays the vehicle image at the position reflecting the actual position of the vehicle in the road image. Thus, in the information processing device, the occupant can grasp the positional relationship between the vehicle and the target object by viewing the display unit.

(16) As described above, in the information processing device according to the present disclosure, it is possible to display, on the in-vehicle display, the contents related to the possibility that the target object will actually cross the road.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the present disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

(2) FIG. 1 is a block diagram showing the hardware configuration of a vehicle;

(3) FIG. 2 is a block diagram showing an example of the functional configuration of the vehicle;

(4) FIG. 3 is a first flowchart showing the flow of control processing;

(5) FIG. 4 is a second flowchart showing the flow of control processing;

(6) FIG. 5 is a first explanatory diagram showing a display example displayed on the display unit while the vehicle is running; and

(7) FIG. 6 is a second explanatory diagram showing a display example displayed on the display unit while the vehicle is running.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) A vehicle **10** according to this embodiment will be described. The vehicle **10** may be an engine vehicle, a hybrid electric vehicle, or a battery electric vehicle, but in the present embodiment, the vehicle **10** is an engine vehicle as an example.

(9) FIG. 1 is a block diagram showing the hardware configuration of the vehicle **10**. As shown in FIG. 1, the vehicle **10** includes a vehicle-mounted device **20**, an Electronic Control Unit (ECU) **30** and an in-vehicle device **40**. The vehicle-mounted device **20** is an example of an “information processing device”.

(10) The vehicle-mounted device **20** includes a Central Processing Unit (CPU) **21**, a Read Only

Memory (ROM) **22**, a Random Access Memory (RAM) **23**, a storage unit **24**, an in-vehicle communication Interface (I/F) **25**, and an input/output I/F **26**, and wireless communication I/F **27**. The CPU **21**, ROM **22**, RAM **23**, storage unit **24**, in-vehicle communication I/F **25**, input/output I/F **26**, and wireless communication I/F **27** are connected via an internal bus **28** so as to be able to communicate with each other.

(11) The CPU **21** is a central processing unit that executes various programs and controls each section. That is, the CPU **21** reads a program from the ROM **22** or the storage unit **24** and executes the program using the RAM **23** as a work area. The CPU **21** performs control of the above components and various arithmetic processing according to programs recorded in the ROM **22** or the storage unit **24**.

(12) The ROM **22** stores various programs and various data. RAM **23** temporarily stores programs or data as a work area.

(13) The storage unit **24** is configured by a storage device such as an embedded Multi Media Card (eMMC) or Universal Flash Storage (UFS), and stores various programs and various data. The storage unit **24** stores an information processing program **24A** for causing the CPU **21** to execute control processing, which will be described later.

(14) The in-vehicle communication I/F **25** is an interface for connecting with the ECU **30**. The interface uses a communication standard based on the CAN protocol. In-vehicle communication I/F **25** is connected to external bus **29**. Although not shown, a plurality of ECUs **30** is provided for each function of the vehicle **10**.

(15) The input/output I/F **26** is an interface for communicating with an in-vehicle device **40** mounted on the vehicle **10**.

(16) The in-vehicle device **40** is various devices mounted on the vehicle **10**. The vehicle **10** includes a detection unit **40A** and a display unit **40B** as an example of the in-vehicle device **40**.

(17) The detection unit **40A** is a detection device capable of detecting the state of the vehicle **10** including inside and outside the vehicle. The detection unit **40A** contains a camera, a vehicle speed sensor, a millimeter wave sensor, and a Global Positioning System (GPS) device as an example.

(18) A camera as the detection unit **40A** is, for example, an imaging device that performs imaging using an imaging element such as a Charge Coupled Device (CCD) image sensor or a Complementary Metal Oxide Semiconductor (CMOS) image sensor. The camera is provided in the front part of the vehicle **10** and images the front of the vehicle. The image captured by the camera is used, for example, to recognize the inter-vehicle distance from a preceding vehicle traveling in front of the vehicle, lanes, and objects present in front of the vehicle. Images captured by the camera are stored in the storage unit **24**. Note that the camera may be configured as an imaging device for other uses such as a drive recorder and an Advanced Driver Assistance System (ADAS). Moreover, the camera may be connected to the vehicle-mounted device **20** via the ECU **30** (for example, camera ECU).

(19) A vehicle speed sensor as the detection unit **40A** is a sensor for detecting the vehicle speed of the vehicle **10**, and is provided, for example, in the wheels.

(20) A millimeter wave sensor as the detection unit **40A** is a sensor for detecting a target object existing in front of the vehicle, that is, in the traveling direction of the vehicle **10**, and is provided at least in the front portion of the vehicle **10**. The millimeter wave sensor is used to detect a target object by transmitting a transmission wave forward of the vehicle **10** and receiving a reflected wave from an object in front. Here, the target object is, for example, a pedestrian, a bicycle, a wall, an oncoming vehicle, or any other object that may pose a danger to the running of the vehicle **10**. Note that the object to be the target object may be registered in the vehicle **10** in advance, or may be registered by input from the passenger. Moreover, it is desirable that the object to be the target object can be updated such as deletion and addition automatically by the vehicle-mounted device **20** or based on the input of the occupant. Hereinafter, in this embodiment, the target object is assumed to be a pedestrian as an example.

(21) A GPS device as the detection unit **40A** is a device that detects the current location of the vehicle **10**. The GPS device includes an antenna (not shown) that receives signals from GPS satellites. The GPS device may be connected to the vehicle-mounted device **20** via a car navigation system connected to the ECU **30** (for example, a multimedia ECU).

(22) The display unit **40B** is an in-vehicle display for displaying an operation suggestion related to the function of the vehicle **10**, an image related to the description of the function, and the like. The display unit **40B** is provided on a meter panel as an example.

(23) A wireless communication I/F **27** is a wireless communication module for communicating with the outside. The wireless communication module uses communication standards such as 5G, LTE, Wi-Fi (registered trademark), for example.

(24) Next, the functional configuration of the vehicle **10** will be described. FIG. 2 is a block diagram showing an example of the functional configuration of the vehicle **10**. As shown in FIG.

(25) As shown in FIG. 2, the CPU **21** of the vehicle-mounted device **20** has an acquisition unit **21A** and a control unit **21B** as functional configurations. Each functional configuration is realized by the CPU **21** reading out the information processing program **24A** stored in the storage unit **24** and executing it.

(26) The acquisition unit **21A** periodically acquires detection information detected by the detection unit **40A** while the vehicle **10** is running. The detection information includes, for example, an image captured by a camera and detection results detected by a vehicle speed sensor, a millimeter wave sensor, and a GPS device.

(27) The control unit **21B** controls the display content of the display unit **40B** based on the detection information acquired by the acquisition unit **21A**. For example, while the vehicle **10** is traveling, based on the detection information, the control unit **21B** can display, on the display unit **40B**, a vehicle image **50** showing the vehicle **10**, a road image **52** showing the road on which the vehicle **10** is traveling, and a target object image **54** showing the target object (see FIGS. 5 and 6).

(28) FIG. 3 is a first flowchart showing the flow of control processing in which the vehicle-mounted device **20** controls the display content of the display unit **40B**. Control processing is performed by the CPU **21** reading out the information processing program **24A** from the storage unit **24**, developing it in the RAM **23**, and executing it. As an example, the control process shown in FIG. 3 is periodically executed while the vehicle **10** is running.

(29) In **S10** shown in FIG. 3, the CPU **21** acquires detection information detected by the detection unit **40A**. Then, the CPU **21** proceeds to **S11**.

(30) At **S11**, the CPU **21** causes the display unit **40B** to display the vehicle image **50** and the road image **52** based on the detection information acquired at **S10**. Specifically, the CPU **21** causes the vehicle image **50** to be displayed at a position reflecting the actual position of the vehicle **10** in the road image **52**. For example, when the vehicle **10** is traveling on the left side of the road, the CPU **21** displays the vehicle image **50** on the left end of the road image **52**. Then, the CPU **21** proceeds to **S12**.

(31) In **S12**, the CPU **21** determines whether or not the display condition is satisfied based on the detection information acquired in **S10**. Here, when the CPU **21** determines that the display condition is satisfied (**S12**: YES), the process proceeds to **S13**. On the other hand, if the CPU **21** does not determine that the display condition is satisfied (**S12**: NO), it ends the control process. In this embodiment, the CPU **21** determines that the display condition is satisfied, when a target object existing in the traveling direction of the vehicle **10** is detected by the detection unit **40A**, and it is assumed that the positional relationship between the vehicle **10** and the target object is in a predetermined state, and the target object crosses the road. Here, when the positional relationship between the vehicle **10** and the target object is in a predetermined state, the relative collision time divided by the relative speed between the vehicle **10** and the target object, the so-called Time to Collision (TTC), is less than or equal to a predetermined value. The CPU **21** calculates the TTC between the vehicle **10** and the target object using the distance and relative speed between the

vehicle **10** and the target object as the detection result of the millimeter wave sensor, which is the detection unit **40A**. Also, when it is assumed that the target object crosses the road, the actual orientation of the target object faces the road. The CPU **21** uses a known image recognition technique to determine the direction of the body of the target object from the image captured by the camera, which is the detection unit **40A**.

(32) In **S13**, the CPU **21** causes the display unit **40B** to display the target object image **54** based on the detection information acquired in **S10**. Specifically, the CPU **21** causes the target object image **54** to be displayed in the road image **52** at a position that reflects the actual position of the target object. For example, when the actual position of the target object is outside the road, the CPU **21** displays the target object image **54** at a position indicating the outside of the road in the road image **52**. Then, the CPU **21** terminates the control process.

(33) FIG. **4** is a second flowchart showing the flow of control processing. As an example, the control process shown in FIG. **4** is periodically executed when the target object image **54** is displayed on the display unit **40B**.

(34) In **S20** shown in FIG. **4**, the CPU **21** acquires detection information detected by the detection unit **40A**. Then, the CPU **21** proceeds to **S21**.

(35) In **S21**, the CPU **21** determines whether the target object is expected to cross the road based on the detection information acquired in **S20**. Here, when the CPU **21** determines that the target object is not expected to cross the road (**S21: YES**), the process proceeds to **S22**. On the other hand, if the CPU **21** does not determine that the target object is not supposed to cross the road (**S21: NO**), it ends the control process. Here, the case where the target object is not supposed to cross the road is the case where the actual orientation of the target object existing outside the road does not face the road. The CPU **21** uses known image recognition technology to determine the position of the target object and the orientation of the body from the image captured by the camera, which is the detection unit **40A**.

(36) In **S22**, the CPU **21** erases the target object image **54** from the display unit **40B**. Then, the CPU **21** terminates the control process.

(37) Next, a display example displayed on the display unit **40B** as a result of the control processing shown in FIG. **3** or **4** being performed by the vehicle-mounted device **20** will be described.

(38) FIG. **5** is a first explanatory diagram showing a display example displayed on the display unit **40B** while the vehicle **10** is running. Specifically, FIG. **5** is a display example of the display unit **40B** when the vehicle **10** is running on a straight road.

(39) The display unit **40B** shown in FIG. **5** displays a vehicle image **50** showing the vehicle **10** traveling in the direction of the arrow **S**, and a road image **52** showing the straight road on which the vehicle **10** travels. As an example, the vehicle image **50** is a substantially rectangular figure. In the road image **52**, the area between the two straight lines is the inside of the road, and the outside of the two straight lines is outside the road.

(40) In addition to the vehicle image **50** and the road image **52**, the display unit **40B** shown in FIG. **5** displays the target object image **54** when the CPU **21** determines that the display condition is satisfied. The target object image **54** reflects the actual position of the target object, and is displayed outside the road on the right side of the vehicle image **50** in the figure on the forward side in the traveling direction of the vehicle **10**.

(41) As an example, the target object image **54** is composed of a circular frame portion **54A** and an icon **54B** displayed within the frame portion **54A** and indicating the type of the target object. In this embodiment, a plurality of types of icons **54B** are provided, and the icon **54B** selected by the CPU **21** based on the acquired detection information is displayed within the frame portion **54A**. In FIG. **5**, an icon **54B** representing a pedestrian is displayed.

(42) Here, the CPU **21** displays a target object image **54** reflecting the actual orientation of the target object. As an example, the display unit **40B** shown in FIG. **5** displays the body of the pedestrian indicated by the icon **54B** such that the body faces the road because the CPU **21**

determined from the detection information that the body of the target object was facing the road (for example, the left side in the drawing).

(43) FIG. 6 is a second explanatory view showing a display example displayed on the display unit **40B** while the vehicle **10** is running. Specifically, FIG. 6 is a display example of the display unit **40B** after a predetermined time has elapsed since the display example shown in FIG. 5 was displayed.

(44) Here, the CPU **21** changes the vehicle image **50** and the road image **52** following changes in the actual position of the vehicle **10**. Accordingly, in FIG. 6, the position of the vehicle image **50** has changed from that in FIG. 5 to the upper side in the drawing when the vehicle **10** travels along a straight road. Although not shown, when the vehicle **10** enters a curved road from a straight road, the CPU **21** changes the two straight lines of the road image **52** into two curved lines.

(45) In addition, the CPU **21** changes the position of the target object image **54** to follow changes in the actual position of the target object. For example, although not shown, when a target object enters the road, the CPU **21** displays the target object image **54** inside the road indicated by the road image **52**.

(46) Here, in FIG. 6, the CPU **21** erases the target object image **54** from the display unit **40B** as it determines that the target object is no longer expected to cross the road. As an example, the CPU **21** erases the target object image **54** because the direction of the body of the target object determined from the image captured by the camera, which is the detection unit **40A**, is not facing the road.

(47) As described above, in the vehicle-mounted device **20**, the CPU **21** displays a target object image **54** in the road image **52** at a position that reflects the actual position of the target object, as a function of the control unit **21B**, when the target object is detected by the detection unit **40A**, and it is assumed that the positional relationship between the vehicle **10** and the target object is in a predetermined state, and the target object crosses the road. As a result, the vehicle-mounted device **20** displays the target object image **54** on the condition that the target object crosses the road, so that the vehicle-mounted device **20** can display, on the in-vehicle display, the content associated with the possibility that the target object crosses the road in reality.

(48) Also, when it is assumed that the target object crosses the road, the actual orientation of the target object faces the road. As a result, the vehicle-mounted device **20** described above can determine the possibility of the target object crossing the road based on the actual orientation of the target object, and control whether to display the target object image **54**.

(49) In the vehicle-mounted device **20**, the CPU **21**, as a function of the control unit **21B**, displays the target object image **54** reflecting the actual orientation of the target object. Accordingly, in the vehicle-mounted device **20**, the occupant can grasp the actual orientation of the target object by looking at the display unit **40B**.

(50) Further, in the vehicle-mounted device **20**, the CPU **21** displays the target object image **54** as a function of the control unit **21B**, and then deletes from the display unit **40B** the target object image **54** to the display unit when the target object is no longer expected to cross the road. As a result, in the vehicle-mounted device **20**, the amount of information to be displayed on the display unit **40B** can be reduced when the target object is no longer expected to cross the road.

(51) In the vehicle-mounted device **20**, the CPU **21** causes the vehicle image **50** to be displayed at a position reflecting the actual position of the vehicle **10** in the road image **52** as a function of the control unit **21B**. Accordingly, in the vehicle-mounted device **20**, the occupant can grasp the positional relationship between the vehicle **10** and the target object by looking at the display unit **40B**.

OTHERS

(52) In the above-described embodiment, the vehicle-mounted device **20** is used as an example of the information processing device, but the present disclosure is not limited to this. An external device such as a server that is not mounted on the vehicle **10** may be used as an example of the

information processing device, and a combination of the external device and the vehicle-mounted device **20** may be used as an example of the information processing device. For example, when the combination of the vehicle-mounted device **20** and an external device is used as an example of an information processing device, at least part of each functional configuration of the CPU **21** of the vehicle-mounted device **20** shown in FIG. **2** may be performed by the CPU of the external device. In this case, the control processing shown in FIG. **3** or FIG. **4** is executed by one processor of the CPU **21** of the vehicle-mounted device **20** or the CPU of the external device, or a combination of a plurality of processors of the CPU **21** of the vehicle-mounted device **20** and the CPU of the external device.

(53) In the above-described embodiment, the detection unit **40A** includes a camera, a vehicle speed sensor, a millimeter wave sensor, and a GPS device, but the configuration of the detection unit **40A** is not limited to this. For example, the detection unit **40A** includes a millimeter wave sensor as a detection device that detects a target object existing in the traveling direction of the vehicle **10**, but may include a laser imaging detection and ranging (LIDAR) instead of or in addition to the millimeter wave sensor, the lidar may be used to detect the target object.

(54) In the above embodiment, the display unit **40B** is provided on the meter panel, but the arrangement of the display unit **40B** inside the vehicle is not limited to this. For example, the display unit **40B** may be provided on an instrument panel, or configured as a HUD that illuminates the display surface of the front windshield.

(55) In the above-described embodiment, the display unit **40B** is a display device provided in the vehicle **10**. However, the display unit **40B** is not limited to this. A mobile terminal such as a smartphone of an occupant may be set in a vehicle, and the mobile terminal may be used as the display unit **40B**.

(56) In the above embodiment, the case where the positional relationship between the vehicle **10** and the target object is in the predetermined state is the case where the TTC between the vehicle **10** and the target object is equal to or less than a predetermined value, but the present disclosure is not limited to this. For example, another case, such as when the distance between the vehicle **10** and the target object is equal to or less than a predetermined value, may correspond to the case where the positional relationship between the vehicle **10** and the target object is in a predetermined state.

(57) In the above embodiment, the case where the target object is assumed to cross the road is the case where the actual direction of the target object faces the road, but the present disclosure is not limited to this. For example, when it is assumed that the target object crosses the road, other cases such as when the target object raises its hand may be used.

(58) In the above embodiment, the actual orientation of the target object to be reflected in the target object image **54** is appropriately determined according to the type of target object. For example, in the above-described embodiment, the target object is a pedestrian, so the orientation of the pedestrian's body is reflected in the target object image **54**. When the target object is a bicycle, the orientation of the tire may be reflected in the target object image **54**.

(59) In the above embodiment, the case where the target object is not expected to cross the road is the case where the actual direction of the target object existing outside the road is not facing the road, but the present disclosure is not limited to this. For example, when the target object is not supposed to cross the road, other cases such as when the target object existing outside the road stays in place for a predetermined time or longer may be used.

(60) Note that the control processing executed by the CPU **21** by reading the software (program) in the above embodiment may be executed by various processors other than the CPU. In this case, the processor is a Programmable Logic Device (PLD) whose circuit configuration can be changed after manufacturing, such as a Field-Programmable Gate Array (FPGA), and an Application Specific Integrated Circuit (ASIC) for executing specific processing. A dedicated electric circuit or the like, which is a processor having a specially designed circuit configuration, is exemplified. Also, the control processing may be executed by one of these various processors, or a combination of two or

more processors of the same or different type (for example, a plurality of FPGAs, a combination of a CPU and an FPGA, etc.). Further, a hardware configuration of the various processors is, more specifically, an electric circuit in which circuit elements such as semiconductor elements are combined.

(61) Further, in the above-described embodiment, the information processing program **24A** is stored (installed) in advance in the storage unit **24**, but the present disclosure is not limited to this. The information processing program **24A** may be provided in a form recorded in a recording medium such as a Compact Disk Read Only Memory (CD-ROM), a Digital Versatile Disk Read Only Memory (DVD-ROM), and a Universal Serial Bus (USB) memory. Also, the information processing program **24A** may be downloaded from an external device via a network.

Claims

1. An information processing device comprising one or more processors configured to: display a target object image showing a target object at a position reflecting an actual position of the target object in a road image showing a road and displayed on a display in a vehicle in a case where i) the target object present in a traveling direction of the vehicle traveling on the road is detected by a detector, ii) a positional relationship between the vehicle and the target object is in a predetermined state, and iii) the target object is expected to cross the road; and erase the target object image from the display in a case where the target object is no longer expected to cross the road after the target object image is displayed.
 2. The information processing device according to claim 1, wherein the target object is expected to cross the road when an actual direction of the target object is a direction toward the road.
 3. The information processing device according to claim 2, wherein the one or more processors are configured to display the target object image reflecting the actual direction of the target object.
 4. The information processing device according to claim 1, wherein the one or more processors are configured to display a vehicle image showing the vehicle at a position reflecting an actual position of the vehicle in the road image.
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